The top graphic in figure 7.18 shows that the target and prediction from Dist-DQN now match almost exactly after training (you may not even be able to see that there are two overlapping distributions anymore). It works! The loss plot on the bottom of figure 7.18 has those spikes from each time the target network is synchronized to the main model and the target distribution suddenly changes, leading to a higher than normal loss at that time step. We can also look at the learned distributions for each action for each sample in the batch. The following listing shows how to do this.

Figure 7.18. Top: The concatenated action-value distributions for all three actions after training. Bottom: Loss plot over training time. The baseline loss is decreasing, but we see ever-increasing spikes.